

Polyfills

A polyfill is:

A piece of code that implements a feature in older environments that do not natively support it.

In simple words:

If the browser/JS engine doesn't support a feature → we write it manually.

Older browsers didn't support:

- `Array.prototype.map`
- `Promise`
- `Object.assign`
- `fetch`
- `Array.prototype.includes`

So developers wrote polyfills to mimic their behavior.

Why Do Polyfills Exist?

Because JavaScript evolves.

The language is standardized by:

- 👉 Ecma International
- 👉 Through the specification: ECMAScript

But browsers implement features at different speeds.

So when a new feature appears:

- Modern Chrome supports it
- Older Safari may not
- Old Internet Explorer definitely doesn't

Polyfills bridge that gap.

Simple Polyfill Example

◆ Array.prototype.includes

```
if (!Array.prototype.includes) {  
  Array.prototype.includes = function(searchElement, fromIndex = 0) {  
    const len = this.length;  
  
    for (let i = fromIndex; i < len; i++) {  
      if (this[i] === searchElement ||  
        (Number.isNaN(this[i]) && Number.isNaN(searchElement))) {  
        return true;  
      }  
    }  
  
    return false;  
  };  
}
```